



Durability of Pultruded GFRP Rebar Subjected to Natural Weathering in Alkaline Solution for 10 Years

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Abstract

Pultruded glass fibre reinforced polymer composite rebar is being considered as an alternate of the traditional material such as steel rebar in the construction industry because of its corrosion resistance property and longevity. In this work, long term mechanical, thermomechanical and chemical resistance properties of the GFRP rebar were investigated after being exposed in alkaline solution for 10 years in the natural weathering conditions. Micro-CT scans revealed that the rebar possessed very little porosity which increased but remained significantly low even after conditioning for 10 years. Low water absorption of the control and the conditioned rebars also supported the finding of small fraction of porosity content. Polymer matrix of the rebar was found to be softened due to plasticisation during conditioning in the alkaline solution as measured through its glass transition temperature (T_g); however, the ultimate T_g was remained almost unchanged. Tensile strength and tensile modulus was found to retain up to 70 and 100%, respectively. Fibre/matrix de-bonding was the main tensile failure mechanism as observed through microscopic analysis. Elemental analysis of the conditioned rebar indicated that the compositions of fibre glass changed very little, which supported the greater retention of tensile modulus after conditioning for 10 years.

Keywords Alkaline solution · Glass fibre polymer composite · Glass transition temperature · Natural weathering · Tensile strength

Introduction

Corrosion of steel within reinforced concrete is one of the major problems faced by the construction industry. To maintain structural integrity of the steel reinforced structures, corrosion protection measures are required [1–3]. The harsh environments not only corrode the steel but also result in concrete cracking, spalling, and delamination, which ultimately trim down the service life [4, 5]. In addition, the high weight to strength also remains a major concern in the field of civil engineering, e.g. concrete structures, anchoring, structural repair, and rehabilitation [6]. To ensure the long term performance of concrete structures, a non-corrosive

composite material such as fibre reinforced polymer (FRP) rebar can be an appealing alternative to the steel rebar [7, 8]. Several types of FRP rebar such as carbon, glass, basalt, and aramid are manufactured and tested for use in the constructions [9–13]. However, the higher capital cost remains a major concern, but less maintenance cost and better performance over the long term have led to an increase of use of FRP reinforced concrete in the construction sector [14]. Glass fibre is the most economical synthetic fibre, thus has been successfully used for FRP rebar in civil structures [15–17].

The durability of FRP is defined as the ability of FRP to resist the thermomechanical, mechanical, and environmental loads such as chemical attack, and physical aging over a long time period. In addition, it is related to the performance of composite materials over a long-time scale measurable in 20 or even 50 years [18]. Many research works investigated the durability of GFRP rebar in different environments either in accelerated conditions or in natural environmental conditions. In an accelerated mode, GFRP rebar is exposed in relevant harsh environment at elevated temperature with or without pre-load. For instance, Sen et al. [19] used alkaline

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solution with a pH between 13.35 and 13.50 in an environment at ambient temperature to expose E-glass/vinyl ester rebar for 1 to 9 months under stressed and unstressed conditions. The researchers recorded that the specimens stressed to 10% lost 70% of their utmost tensile strength while under unstressed conditions the specimens lost 63% after 9 months of exposure. In reference [20], the tensile strength retention of GFRP rebars made of various resins (e.g. polyester and vinyl ester resin) and reinforcements (e.g. electric resistance glass, electric and corrosion resistance glass, and alkali resistance glass) exposed in high pH solution and concrete was discussed. The tensile strength of alkali resistant glass/vinyl ester rebar was shown to be unaffected by a 30% stress level and 140 days of ageing. The tensile strength of E-glass/vinyl ester, on the other hand, reduced by 16–17%. Jin et al. [21] investigated the tensile characteristics of GFRP rebars subjected to cyclic tension at various stress levels and submerged in alkaline solution for a couple of days to assess the safety and serviceability of GFRP reinforced structures. The findings demonstrate that the tensile strength and rate of degradation of GFRP rebars are mostly influenced by the alkaline solution and stress levels. Foruzanmehr et al. [22] studied the chemical resistance of polyester and vinyl ester based GFRP dowels for concrete pavement in alkaline and saline solutions simulated environments. The vinyl ester based GFRP dowels showed stable mechanical properties as compared to polyester based GFRP on exposure to simulated environments. The effect of hydrothermal aging on the properties of GFRP rebars of isophthalic polyester or vinylester resin with E-glass reported in the reference [23]. The GFRP rebars were immersed in an alkaline solution (pH 12.6) for 1000 h at 23 and 60 °C. The results indicated that the rebars of vinylester resin had greater chemical resistance than the rebars of polyester resin. Due to alkaline ageing, tensile strength loss of vinyl ester rebar was found to be 3.8 and 5.5% at 23 and 60 °C.

Investigations on the durability of GFRP rebar after exposure to natural environmental conditions are also carried out by several researchers. For example, GFRP rebar in concrete showed a limited degradation even after 8 years of exposure to harsh natural environmental conditions including freezing, defrosting, and de-icing salts [24]. Gooranorimi and Nanni [16] evaluated the performance of GFRP with vinyl ester resin reinforced concrete samples from a bridge after 15 years of service. The scanning electron microscopy did not detect any decline in fibre-matrix adhesion, while variation in chemical composition was not observed by energy dispersive X-ray spectroscopy. Moreover, T_g and glass fibre content were comparable to pristine values. Similarly, the durability of GFRP rebar removed from different bridges after 15 to 20 years of exposure to harsh natural environmental conditions was evaluated by Al-Khafaji and co-workers [25]. The result showed very little damage to the GFRP rebar

even after 20 years of service in harsh environmental conditions. Starkova et al. [26] studied the radial and axial diffusivities of four types of commercially available FRP rebars of carbon, glass, and aramid fibres exposed to moisture for 15 years. They concluded that the exposure of FRP rebars in a humid environment decreases their interlaminar shear strength by 10 to 50%. Ramanathan and co-workers [14] studied the 18 years in service dry dock concrete and GFRP repaired rebar samples from Pearl Harbor in Oahu, Hawaii. The concrete showed clear evidence of damage occurring over the years but the GFRP core remained unaffected as compared to the exposed surface which had minor damage.

Even though a number of investigations have been reported on the durability of GFRP rebars exposed to different harsh environments, the important properties such as tensile strength and tensile modulus of the GFRP rebars conditioned for long period have not been well documented. Majority of the existing literature results were based on conditioning at high temperatures for short period, where high temperature facilitated faster diffusion of alkaline solution in to the matrix and reaching to fibre/matrix interfaces. Also high temperature could cause internal crack and damage to the matrix due to post curing, which allows too much diffusion of alkaline solution or water in to the rebar in a short period. In most of the reports, the GFRP rebars were conditioned for 3 to 36 months in accelerated mode at elevated temperatures, and the results represented a “worst” case scenario of the its performances. Because of lack of long term performance data especially the tensile property, there is still a reluctance to use GFRP rebar widely in civil structures. This paper presented systematic investigations and long term tensile property of a pultruded GFRP rebar exposed in natural environmental and alkaline solution for 10 years.

Experimental

Material

The spiralled GFRP rebars of 13.2 mm core diameter were made through pultrusion process using boron-free ECR glass and Epoxy vinyl ester resin. In the manufacturing process, the glass fibre was impregnated in resin mixture and pulled through a hot steel die for consolidation and polymerisation reaction. The pultruded round rod was machined for the spiral texture.

Alkali Resistance Test

In one litre of deionized water, 118.5 g of Ca(OH)_2 , 0.9 g of NaOH, and 4.2 g of KOH were mixed to make an alkaline solution [27]. pH of the alkaline solution at the start was about 13 (a representative pH value of mature concrete

Fig. 1 Photographs of the control and conditioned GFRP rebars **a** top view and **b** side view



pore solution), and the final pH was found to be around 9.3. The GFRP rebars were fully immersed in a plastic container filled with the alkaline solution. The container was then exposed to outdoor natural weathering conditions for 10 years. The local annual average outdoor temperature was about 13.9 °C, where the temperature variation was about 9.5 °C. Photo of the control and 10 years alkaline solution exposed (10-years) rebars is shown in Fig. 1. The alkaline precipitates i.e. mainly the slurry of $\text{Ca}(\text{OH})_2$ on the rebars were carefully scrubbed prior to the testing of the rebar.

Micro-CT Scan and Porosity Measurement

The specimens were scanned using Bruker Skyscan 1275 operated at 40 kV 100 μA without filter. The scanning parameters were 0.4° rotation step, 360° scanning and 4 frame averaging. Total number of shadow projections (raw images) was 901. Camera settings were 1944 $\times$ 1536 resolution and 67 ms exposure. NRecon program was used to calculate the cross sections from raw images. Then CTan software was applied to binarize images to segregate areas of different density. Porosity was calculated from the binarized images. Figure 2 presents (a) Micro-CT test chamber, (b) NRecon image (Longitudinal) and (c) NRecon image (cross-section).

Water Absorption Test

The control and 10-years rebar specimens of length 40 mm was first conditions at 23 °C for 48 h in the laboratory environment. The specimens were then weighed on a digital scale with accuracy of ± 0.0000 g. Afterwards,

the specimens were placed in a glass container filled with deionized water at 60 °C to measure the water absorption. Weight of the specimens was measured in different time intervals for up to 4488 h. The surface moisture of the specimens was carefully wiped off each time before weighing.

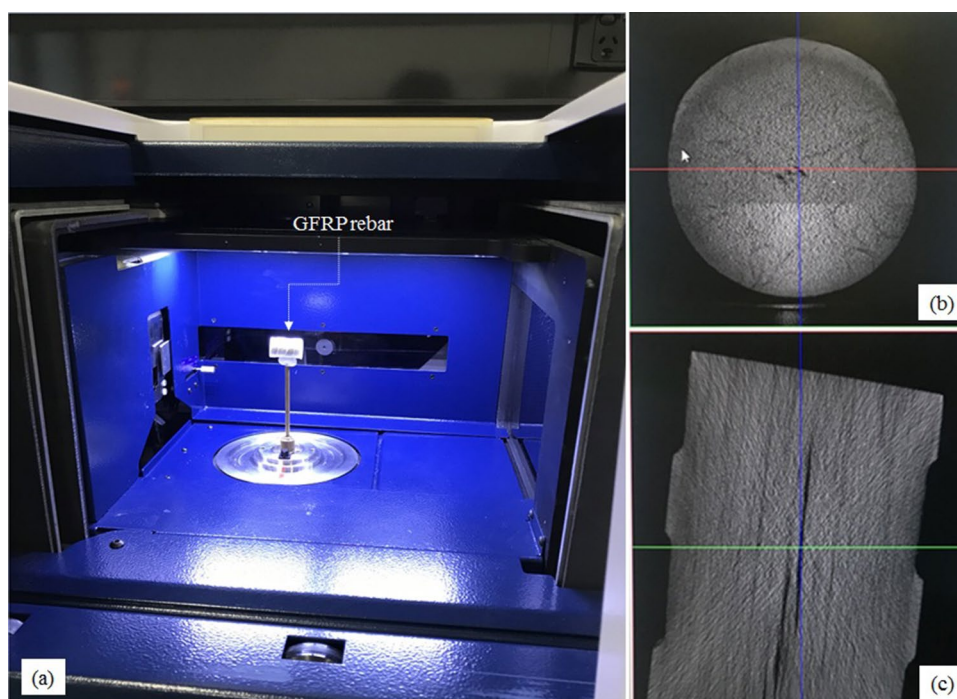
Dynamic Mechanical Thermal Analysis

In three point bending mode, a dynamic mechanical thermal analyser (DMTA V: Rheometric Scientific Inc.) was used to conduct the thermodynamic analysis. The analyzer was heated at a rate of 3 °C/min at 1 Hz from 25 to 190 °C. Within the linear viscoelastic area, the strain amplitude was 0.05%. The test items were rectangular coupons measuring approximately 40 $\times$ 5 $\times$ 0.5 mm³. From each type of material, three specimens were tested. Each specimen was tested twice, once for the first scan and again for the second scan.

Tensile Test

Tensile test specimens were made using the parameters outlined in ASTM D7205/D7205M, Standard Test Method for Tensile Properties of Fiber Reinforced Polymer Matrix Composite Bars. Two ends of the rebar were fixed in epoxy resin-filled steel plugs. A universal testing machine (WDW-200E) fitted with a 200 kN load cell was used to perform the test on six duplicates of each type of material. The cross-head speed was 5 mm/min.

Fig. 2 Photographs of **a** Micro-CT test chamber, **b** NRecon image of longitudinal section of the rebar and **c** NRecon image of cross-section of the rebar



Energy Dispersive X-ray Spectroscopy (EDS)

Energy dispersive X-ray spectroscopy analysis was conducted using a JEOL 7200 SEM machine, and the EDS map data was analysed by AZtec software. The specimens were carbon coated prior to the test.

Scanning Electron Microscope (SEM)

The samples were analysed using Hitachi Regulus 8230 High-Resolution scanning Electron Microscope at 3 kV with a secondary electron detector. Prior to analysis, the samples were metalized with platinum in a Quorum Q150V plus sputter coater.

Results and Discussion

Porosity

In a material, porosity refers to the void fraction and the characterization of their connections. An imperfect manufacturing process could lead to the formation of voids, micro-cracks, or fibre-rich zones in the rebar [20]. Voids could be formed through the inhomogeneous velocity and infiltration of liquid resin into the glass fibre roving bundles, and from the gas released during resin cure reactions. These voids could facilitate diffusion of water and alkaline ions at the vicinity of fibre/matrix interface during the ageing period, which then could lead corrosion of the glass fibre and faster

Table 1 Percent porosity of the control and conditioned (10-years) GFRP rebars

GFRP rebar orientation inside Micro-CT chamber	Porosity (%)	
	Control	10-years
Longitudinal	0.0013	0.0368
Cross-section	0.1272	0.1007

deterioration of the physical and mechanical properties of the rebar.

Table 1 presents the summary of porosity of the control and 10-years rebars as measured through micro-CT scan. In general, % porosity of the control rebar was found to be very small in both orientations (Fig. 2b, c) where the majority of porosity was measured in the orientation of cross-section (Fig. 2c), i.e. the transverse direction of resin flow. Other study also reported the presence of high void content when measured in the transverse direction of resin flow compared to axial flow of resin [28]. Again in Table 1, % porosity in the longitudinal orientation increased from 0.0013 to 0.0368 after 10 years of ageing in alkaline solution. This increase in porosity could be due to fibre/matrix de-bonding as a result of diffusion of water. In contrast, % porosity in the cross-sectional orientation was not increased even after the ageing, which could be due to less fibre/matrix de-bonding occurred as a result of little diffusion of water, because pores near to the spiralled surface of the rebar were filled with alkaline precipitates as can be observed in Fig. 1. Wang et al. [7] reported the porosity of helically wrapped GFRP rebar as

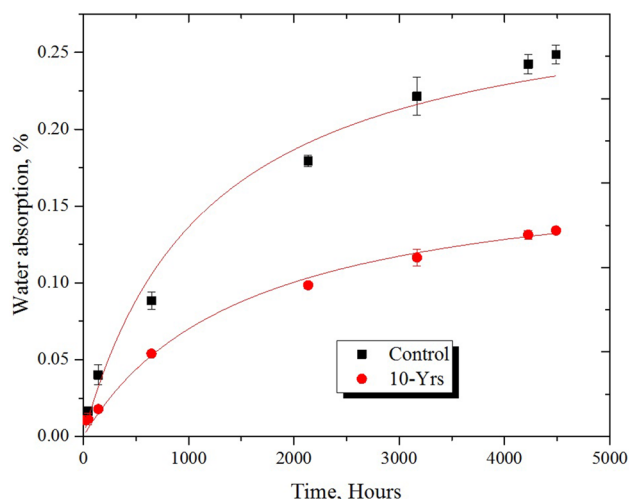


Fig. 3 Water absorption of the control and conditioned GFRP rebars with respect to the immersion time in deionized water at 60 °C

measured through CT-scan technique. They found that the porosity of control rebar was increased from 0.3 to 1.4 after ageing in simulated seawater at elevated temperature. When compared with the present work results, considerably high porosity results of the other work [7] was thought to be due to the difference in the manufacturing process of the rebar.

Water Absorption

Characterisation of water absorption is one of the efficient ways to monitor durability performance of GFRP rebar. Figure 3 presents water absorption results of the control and 10-years rebars with respect to the immersion time. After the initial fast diffusion, water absorption continued to increase gradually with time, and did not reach to an equilibrium even after 6 months of testing at 60 °C. This non-Fickian [29] water absorption behaviour was ascribed to capillary transport of water gradually to the fibre/matrix interface during the water absorption testing period [30]. Similar trend in water absorption also reported in other studies [31, 32], which demonstrated that water absorption in fibre reinforced composite was not stabilized even after immersion for long period.

It can also be observed in Fig. 3 that the water absorption of 10-years rebar was less than the control rebar. This could be a result of decrease of the channels to diffuse water into the fibre/matrix interface because of the deposit of alkaline precipitates on the 10-years rebar as seen in Fig. 1. The average water absorptions of the control and 10-years rebars were found to be 0.2486 and 0.1341%, respectively, after 4488 h immersion at 60 °C. The water absorption of the control and 10-years rebars were found to be less than the other results as reported in the literatures [5, 14, 33].

Dynamic Mechanical Thermal Analysis (DMTA) and Glass Transition Temperature (T_g)

T_g , the temperature at which a polymer is transformed into a rubbery solid from a glassy solid, is a critical metric for determining the degradation and stability of the polymer matrix of GFRP rebar subjected to a hostile environment. T_g is dependent upon the chain flexibility and free volume associated with the chemical structure of polymer matrix. DMTA is a powerful technique to measure the T_g of polymer matrix of GFRP rebar. From DMTA traces, various glass transitions of the polymer matrix can be derived from (i) the measure of the ultimate energy composed in the material after a complete cycle of oscillation (storage modulus), (ii) loss modulus which is assumed to be equivalent to the energy amount dissipated as heat by the specimen and (iii) mechanical damping factor that refers to the ration of the loss modulus to storage modulus. This is associated with the polymer chains' molecular mobility degree.

Typical DMTA curves and glass transitions of the GFRP rebar are presented in Fig. 4. In these curves, intersect and onset point temperatures were considered as T_g [34–36]. As can be seen in Fig. 4a, storage modulus (E') of the GFRP rebar in the glassy state was approximately stable up to the glass transition temperature ($T_{g(E'-onset)}$) and then a rapid declining trend was observed when the temperature went through above $T_{g(E'-onset)}$ in the rubbery state. $T_{g(E'-onset)}$ is useful in the design and application of GFRP rebar as it describes the absolute temperature limit for structural properties to remain stable. Again in Fig. 4a, loss modulus (E'') was nearly steady below the glass transition temperature ($T_{g(E''-onset)}$) but raised to a pronounced peak at the onset temperature, which corresponds to the maximum heat dissipation per unit deformation. $T_{g(E''-onset)}$ indicates viscoelastic property of polymer matrix, and $T_{g(E''-peak)}$ at the peak of E'' describes degradation and stability.

In Fig. 4b, like E'' , damping factor ($\tan\delta$) was steady below the glass transition temperature ($T_{g(\delta-onset)}$) but increased to a pronounced peak at the onset temperature, which corresponds to high damping due to the initiation of motion in segments of the main polymer chain as said earlier. In addition, the temperature at which the $\tan\delta$ peak occurred is commonly known as glass transition temperature ($T_{g(\delta-peak)}$) and generally easier to identify as a discrete temperature.

The summary of different glass transitions of the control and 10-years GFRP rebars is presented in Table 2. The average $T_{g1(E'-onset)}$ of the control rebar was measured as 117.3 °C, which was found to decrease to 114.2 °C after 10 years of ageing in alkaline solution. This small drop of $T_{g1(E'-onset)}$ by 3.1 °C was attributed to softening i.e. plasticisation of the polymer matrix by the absorbed water during ageing. After the 2nd scan, $T_{g2(E'-onset)}$ of the control rebar

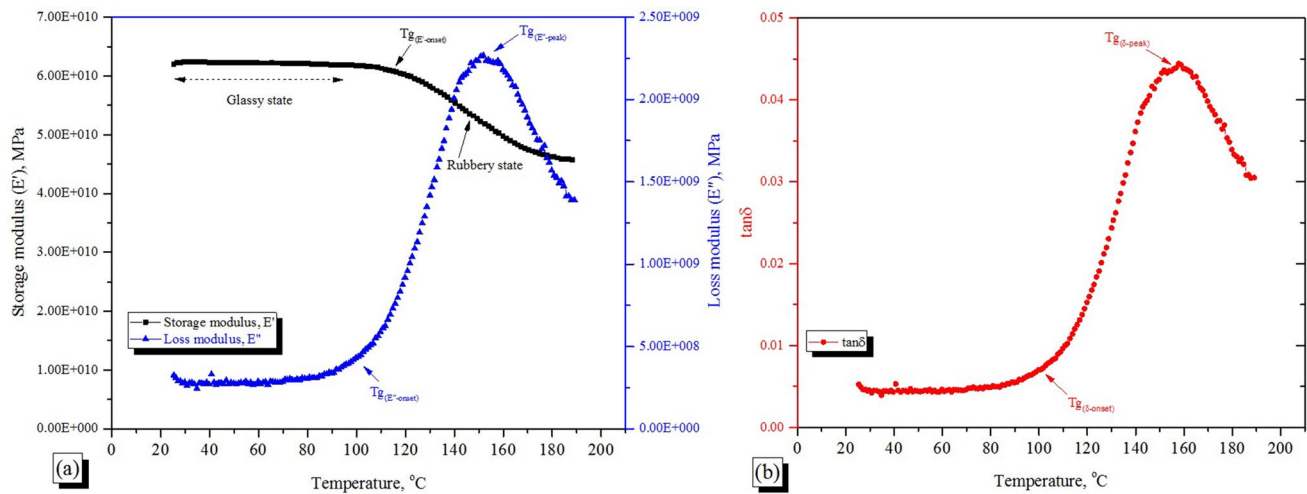


Fig. 4 Typical DMTA thermographs **a** storage and loss modulus with respect temperature and **b** mechanical damping factor, $\tan\delta$, with respect to temperature

Table 2 Different glass transition temperatures of the control and conditioned (10-years) GFRP rebars as measured by DMTA

Glass transitions	Control rebar °C		10-Years rebar °C		Statistical analysis of paired sample t-test of the 2nd scans of control and 10-years rebars
	1st scan T_{g1}	2nd scan T_{g2}	1st scan T_{g1}	2nd scan T_{g2}	
$T_{g(E'-onset)}$	117.3 (± 3.3)	127.2 (± 1.6)	114.2 (± 4.9)	127.6 (± 1.9)	At the 0.05 level, the difference of the population means is not significantly different from the test difference.
$T_{g(E''-onset)}$	107.2 (± 3.4)	111.9 (± 1.3)	98.1 (± 3.1)	112.4 (± 1.7)	At the 0.05 level, the difference of the population means is not significantly different from the test difference.
$T_{g(E''-peak)}$	148.4 (± 4.2)	158.0 (± 4.9)	149.1 (± 3.0)	159.5 (± 1.5)	At the 0.05 level, the difference of the population means is not significantly different from the test difference.
$T_{g(\delta-onset)}$	107.7 (± 1.9)	114.3 (± 1.0)	102.9 (± 3.3)	114.2 (± 1.9)	At the 0.05 level, the difference of the population means is not significantly different from the test difference.
$T_{g(\delta-peak)}$	155.7 (± 5.6)	161.6 (± 2.4)	152.8 (± 3.8)	161.6 (± 1.6)	At the 0.05 level, the difference of the population means is not significantly different from the test difference.

was increased to 127.2 °C which was due to anti-plasticization as a result of loss of non-crosslinked and volatile residues e.g. mould release agent during DMTA scan at high temperature. Like the control rebar, $T_{g2(E'-onset)}$ of the 10-years rebar similarly increased to 127.6 °C after the 2nd scan. So $T_{g2(E'-onset)}$ of both the control and 10-years rebars were found to be almost the same, which confirmed that the drop of $T_{g1(E'-onset)}$ by 3.1 °C after the ageing was a reversible change of the polymer matrix caused by the absorbed water. Similar to $T_{g1(E'-onset)}$ of the control rebar, $T_{g1(E''-onset)}$ of the control rebar decreased from 107.2 to 98.1 °C due to ageing. This large drop of $T_{g1(E''-onset)}$ by 9.1 °C suggesting that the absorbed water affected the viscoelastic property

of the polymer matrix significantly. However, after the 2nd scans, $T_{g1(E''-onset)}$ of both the control and 10-years rebars were found to be very close, which again confirmed a reversible chemical change of the polymer matrix occurred during ageing. As can be seen from the 1st scans, $T_{g1(E''-peak)}$ of both the control and 10-years rebars were appeared to be almost similar, which indicated superior stability of the rebar in the alkaline environmental. After the ageing, both $T_{g(\delta-onset)}$ and $T_{g(\delta-peak)}$ decreased by 4.8 and 2.9 °C, respectively as obtained from the first scans. But like the other glass transitions, the loss of $T_{g(\delta-onset)}$ and $T_{g(\delta-peak)}$ appeared to be fully recovered as can be seen from the 2nd scans results. To assess the significance of the difference in T_g between

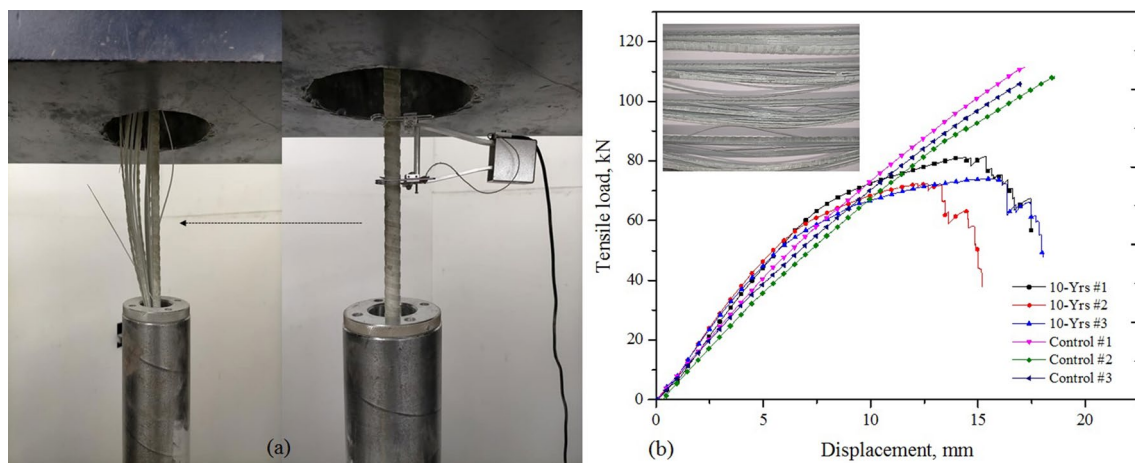


Fig. 5 **a** Photographs of the tensile test and **b** tensile load versus displacement curves and the tensile splintered GFRP rebars

the control and 10-years rebars after the 2nd scans, the data was statistically analysed using paired sample t-test at a confidence level of 95%. As can be seen in Table 1, statistical analysis did not support any significant difference in T_g of the control and 10-years rebars. Overall, different T_g of the rebar was preliminarily dropped (except $T_{g(E''-peak)}$) in the range of 2.9 to 9.1 due to plasticisation by the absorbed water, which was a reversible chemical change as confirmed from the 2nd the scans. The stable value of $T_{g(E''-peak)}$ before and after ageing indicated that the polymer matrix was not degraded during alkaline ageing for 10 years.

Tensile Strength

Typical tensile failure of the rebar is shown in Fig. 5a. As can be observed, the rebar splintered over the free length of the specimen due to a combination of longitudinal extension and transverse compression. The maximum stresses was occurred in the outer fibres and the failure was dominated by the properties of fibre/matrix interface. In Fig. 5b, both the control and 10-years rebars were deformed linearly at the beginning, followed by an amount of non-linear deformation before reaching the maximum load. For the 10-years rebars, once the maximum load was attained, the recorded load diminished gradually as a result of shear failure due to (i) softening of polymer matrix and (ii) deterioration of fibre/matrix interface as a consequence of water diffusion during the long period ageing.

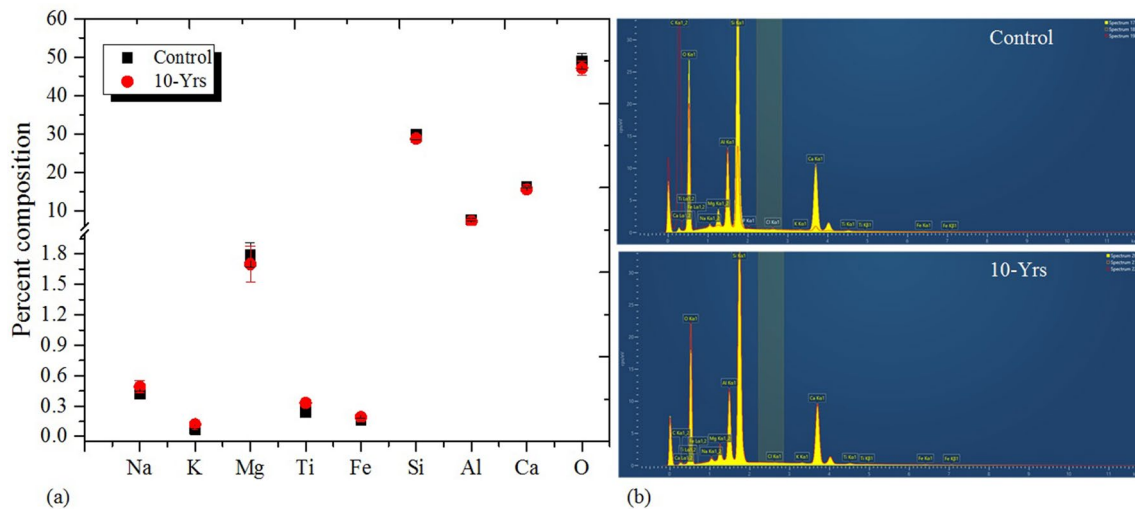
Tensile strength and tensile modulus retentions are presented in Table 3 along with the results of the previous works presented in the references [30, 37] on the rebars from same batch for comparison. For the control rebar, tensile modulus and tensile strength were found to be 55 GPa and 886 MPa, respectively. As can be seen, tensile modulus of the rebar was retained up to 100% after 10 years of natural

weathering. Superior stability of the tensile modulus of rebar was ascribed to the corrosion resistance property of ECR glass fibre [12, 20, 38, 39] as the modulus of the rebar is predominantly depends on the modulus of glass fibre. Other researchers also have reported stability of tensile modulus of GFRP rebar with ECR glass in the alkaline environment [33, 40, 41]. On the other hand, tensile modulus of the rebar of E-glass appeared to be affected after alkaline conditioning as reported in the references [3, 42, 43]. Tensile modulus of the rebar of E-glass also reported to be changed very little in the alkaline environment [44–47], which could be due to good protection of the glass fibre by the polymer matrix from alkaline attack.

In the alkaline media, chemical degradation starts with a nucleophilic attack by hydroxyl ions on the silicon atoms of siloxane bond ($\equiv\text{Si}-\text{O}-\text{Si}\equiv$) of a bare surface of glass fibre to form new groups ($\equiv\text{Si}-\text{OH}$, $\equiv\text{Si}-\text{O}-\text{Na}$, and $\equiv\text{Si}-\text{O}-\text{K}$). Due to the removal of monovalent cations (e.g. Na^+ , K^+) from the glass fibre by forming $\equiv\text{Si}-\text{O}-\text{Na}$ and $\equiv\text{Si}-\text{O}-\text{K}$ groups, the hydroxylated surface (i.e. $\equiv\text{Si}-\text{OH}$) leads to dissolve glass fibre in the alkaline solution. As can be seen from the elemental analysis in Fig. 6a, compositions of the glass fibre changed superficially which indicated that glass fibres of the rebar were chemically stable against the alkaline corrosion, and also the polymer matrix of the rebar protected the glass fibre from alkaline attack; accordingly, the tensile modulus of the rebar remained almost unaffected in the alkaline solution even after 10 years of ageing. It can also be observed in Table 3 that the tensile modulus of the rebar of ECR glass didn't decrease in the alkaline solution even at high temperature after 24 months of ageing as reported earlier in the reference [30]. From this comparison, it can be said that the alkaline ageing time and temperature did not affect the tensile modulus of GFRP rebar. In a recent work [48], tensile modulus of GFRP rebar was found to be not

Table 3 Tensile strength and tensile modulus retention of the GFRP rebars immersed in alkaline solution at different environmental conditions

Property	Pre-sent work tensile test data Test#	Retention in alkaline solution, %				
		Control	10-years	Present work 10 years of natural weathering at average 13.9 °C	Reference [37] 23 months at 23 °C	Reference [30] 24 months at 60 °C
Tensile modulus, GPa	1	53.3	53.5	100	n/a	104
	2	57.8	57.6			
	3	56.3	56.8			
	4	56.0	55.1			
	5	53.5	53.2			
	6	53.3	54.1			
Average		55.0	55.1			
Tensile strength, MPa	1	917	670	70	91	74
	2	907	591			
	3	869	610			
	4	862	570			
	5	906	663			
	6	854	611			
Average		886	619			

**Fig. 6** **a** Percent composition of elements of glass fibre of the control and conditioned GFRP rebars and **b** Energy dispersive X-ray spectra

affected much by the exposure time, environmental conditioning and sustained loads.

Again in Table 3, tensile strength of the rebar was found to retain up to 70%. The loss of tensile strength of the rebar after 10 years of natural weathering in alkaline solution was attributed to plasticising of the matrix and weakening of the fibre/matrix interface as a result of diffusion of water during the long term ageing [49]. The SEM fractography provided characteristic features of the tensile failure of 10-years

rebar. As can be seen in Fig. 7, there was no displacement of the fibres and the failure occurred through splitting in the matrix regions. A well-bonded structure of the fibre/matrix interface was clearly evident in Fig. 8; also the “riverlines” [50] originated from fibre/matrix interface indicated gradual crack propagation during the tensile failure. The characteristic sign of the matrix shear fracture in the form of “cusps” [50, 51] can be observed in Fig. 9. These thicker cusps indicated a greater deformation of the matrix as a result of

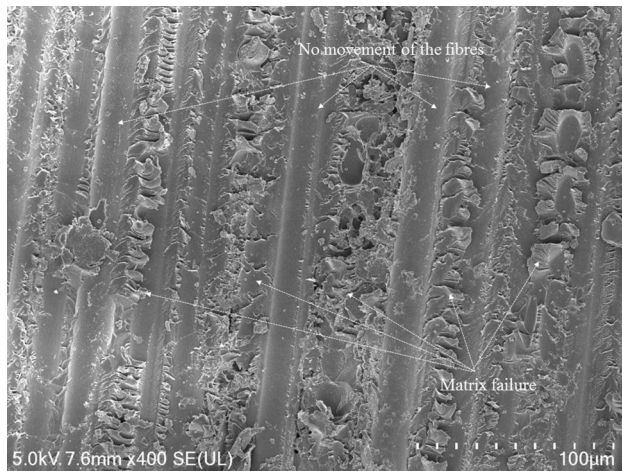


Fig. 7 SEM fractography of the tensile failure of 10-years rebar

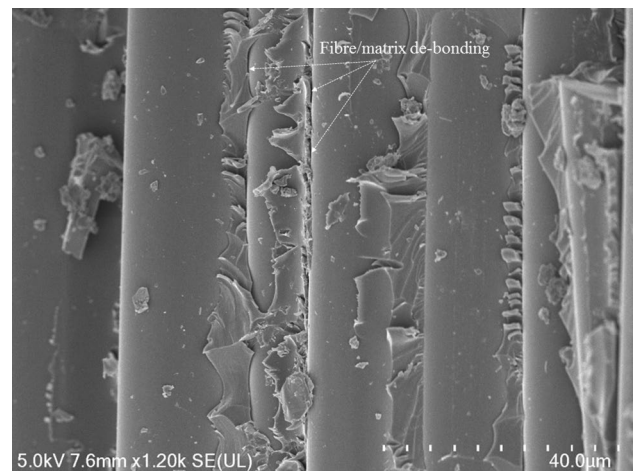


Fig. 10 SEM photograph showing fibre/matrix de-bonding at the interfaces

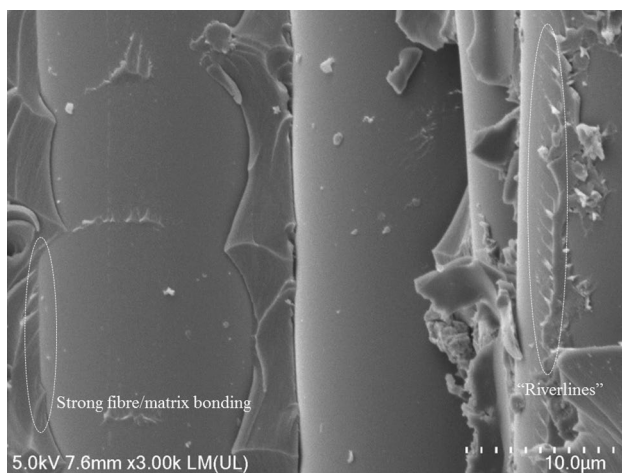


Fig. 8 SEM photograph showing strong fibre/matrix bonding and the “riverlines”

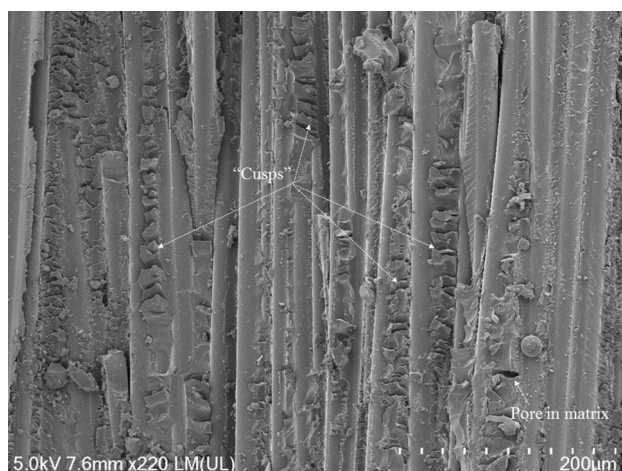


Fig. 9 SEM photograph of matrix failure in the form of “cusps”

higher plasticity of the soften matrix of 10-years rebar [50]. Also in Fig. 9, the presence of pore in the matrix was identified as measured through micro-CT scan. The elastic properties of the matrix and the fibre were different, so de-bonding occurred prematurely at the interfaces between soften matrix and fibre (Fig. 10) as a result of internal stresses build-up during the tensile stress transfer in the longitudinal direction.

Tensile data from the same batch of rebars conditioned for 23 months at 23 °C was presented in the previous report [37] as specified in Table 3. Although the average conditioning temperature of the present work was 13.9, but in the both cases the rebars were aged in non-accelerated conditions in the alkaline solution. Because of approximate similarity in the ageing conditions of the same batch of rebars, the tensile data in the reference [37] could approximately be considered as “intermediate transition period” data. Again, when compared the tensile strength retention of 10-years rebar with that of the reference [30] in Table 3, it can be observed that the tensile strength retention was reached to a plateau in the range of 70 to 74%, and the temperature accelerated the ageing process of GFRP rebar in the alkaline solution [30]. Wu et al. [48] also reported about 70.6 to 63.1% tensile strength retention of the GFRP rebar embedded in concrete beam which was immersed in alkaline solution for eight years.

Furthermore, the tensile strength of 10-years rebar was documented to be greater than the “design tensile strength” as described by ACI 440.1R-15 in Guide for the “Design and Construction of Structural Concrete Reinforced with Fiber-Reinforced Polymer (FRP)” Bars [52]. The design tensile strength can be calculated using the following equations:

$$f_{fu} = C_E f_{fu}^* \quad (1)$$

$$f_{fu}^* = f_{u,ave} - 3\sigma \quad (2)$$

where f_{fu} is the design tensile strength, C_E is the environmental reduction factor, f_{fu}^* is the guaranteed tensile strength, $f_{u,ave}$ is the average tensile strength and σ is the standard deviation. C_E is 0.7 for the GFRP rebar [50]. In this work, $f_{u,ave}$ and σ of the control rebar were 886 and 27.2 MPa, respectively; f_{fu}^* was then calculated according to Eq. 2. Finally, f_{fu} was estimated to 563 MPa using Eq. 1. The ratio of retained tensile strength to design tensile strength was found to be 1.099. Therefore, the retained tensile strength was about 110% higher than the design tensile strength even after conditioning in alkaline solution for 10 years in natural environment.

Conclusion

To date, many studies have been carried out with the primary goal of determining the influence of the alkaline solution or the concrete environment on the characteristics of GFRP rebar in accelerated conditions, i.e. conditioning at elevated temperature for short period; however, the effect of very long term natural weathering in alkaline solution on the properties of GFRP rebar is not well documented. This paper presented the effect of natural weathering for 10 years in alkaline solution on the mechanical, thermomechanical and chemical resistance properties of GFRP rebar. In this work, low water absorption was found to be allied with low porosity content of the GFRP rebar. The moisture absorption of the conditioned rebar was measured to be less than the control rebar because of deposition of alkaline precipitates on the rebar during the long term conditioning. It was also revealed that porosity content depends on the direction of resin flow during the manufacturing of GFRP rebar in pultrusion. Porosity content appeared to be high in the transverse direction of resin flow. A reversible chemical change of the polymer matrix of conditioned rebar was found to be occurred as analysed through the glass transition temperature. Elemental analysis of the rebar indicated that the compositions of ECR fibre glass changed very little, which supported the superior retention (100%) of tensile modulus after conditioning for 10 years. Tensile strength was found to retain above the design tensile strength even after 10 years of immersion in the alkaline solution in natural weathering conditions. The greater stability in tensile strength was believed to be a result of slow and little diffusion of water at the fibre/matrix interface during the conditioning period due to very little porosity content. It was demonstrated that the porosity is an important property of GFRP rebar, which could direct the durability behaviour of GFRP rebar in long term applications.

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Data Availability Different experimental data used in this study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this paper.

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